

Cloud Assignment 3

Group 1

Setting up
vpc
(vpc-0c66d8e
621ff98a01 /
project-vpc)

Create VPC [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances. Mouse over a resource to highlight the related resources.

VPC settings [Info](#)

Resources to create [Info](#)
Create only the VPC resource or the VPC and other networking resources.

☐ VPC only ☒ VPC and more

Name tag auto-generation [Info](#)
Enter a value for the Name tag. This value will be used to auto-generate Name tags for all resources in the VPC.

☒ Auto-generate

IPv4 CIDR block [Info](#)
Determine the starting IP and the size of your VPC using CIDR notation.

65,536 IPs

CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)

☒ No IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block

Tenancy [Info](#)

Number of Availability Zones (AZs) [Info](#)

Preview

VPC [Show details](#)
Your AWS virtual network
project-vpc

Subnets (4)
Subnets within this VPC

- us-east-1a**
 - project-subnet-public1-us-east-1a
 - project-subnet-private1-us-east-1a
- us-east-1b**
 - project-subnet-public2-us-east-1b
 - project-subnet-private2-us-east-1b

Route
Route

Create VPC [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances. Mouse over a resource to highlight the related resources.

Customize subnets CIDR blocks

NAT gateways (\$) [Info](#)
Choose the number of Availability Zones (AZs) in which to create NAT gateways. Note that there is a charge for each NAT gateway.

☐ None ☒ In 1 AZ ☐ 1 per AZ

VPC endpoints [Info](#)
Endpoints can help reduce NAT gateway charges and improve security by accessing S3 directly from the VPC. By default, full access policy is used. You can customize this policy at any time.

☐ None ☒ S3 Gateway

DNS options [Info](#)

☒ Enable DNS hostnames
☒ Enable DNS resolution

Additional tags
Add tags to the VPC and all resources within the VPC. Do not set the Name tag here. Set the Name tag under Name tag auto-generation above or directly in the visualizer.

You can add 49 more tags.

Preview

VPC [Show details](#)
Your AWS virtual network
project-vpc

Subnets (4)
Subnets within this VPC

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 - project-subnet-public1-us-east-1a
 - project-subnet-private1-us-east-1a
- us-east-1b**
 - project-subnet-public2-us-east-1b
 - project-subnet-private2-us-east-1b

Route
Route

Creating my-sql SG

EC2

Dashboard

EC2 Global View

Events

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Flactor IPsec

Security group (sg-0cac603cbe39c9d91 | my-sql) was created successfully

Details

sg-0cac603cbe39c9d91 - my-sql

Details

Security group name

my-sql

Owner

989367328520

Security group ID

sg-0cac603cbe39c9d91

Inbound rules count

1 Permission entry

Description

allows ssh connection

Outbound rules count

1 Permission entry

VPC ID

vpc-0c66d8e621ff98a01

Inbound rules

Outbound rules

Sharing - new

VPC associations - new

Tags

Inbound rules (1)

Search

Manage tags

Edit inbound rules

Name	Security group rule ID	IP version	Type	Protocol	Port range
-	sgr-0a583e13223778105	IPv4	SSH	TCP	22

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Sharing - new

VPC associations - new

Tags

Inbound rules (1)

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Manage tags

Edit inbound rules

Name	Security group rule ID	IP version	Type	Protocol	Port range
-	sgr-0a583e13223778105	IPv4	SSH	TCP	22

Creating SG
my-sql-server

EC2

Dashboard

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Network & Security

Security Groups

Elastic IPs

Security group (sg-0e41b7998d03e4d4b | my-sql-server) was created successfully

Details

sg-0e41b7998d03e4d4b - my-sql-server

Details

Security group name

my-sql-server

Owner

989367328520

Security group ID

sg-0e41b7998d03e4d4b

Inbound rules count

2 Permission entries

Description

allow access to mysql

Outbound rules count

1 Permission entry

VPC ID

vpc-0c66d8e621ff98a01

Inbound rules

Outbound rules

Sharing - new

VPC associations - new

Tags

Inbound rules (2)

Manage tags

Edit inbound rules

Name	Security group rule ID	IP version	Type	Protocol	Port range
-	sgr-028211e4215f39b26	IPv4	SSH	TCP	22
-	sgr-0a80ebe8420b3b010	IPv4	MYSQL/Aurora	TCP	3306

Creating
public
ec2
with
my-sql
SG

EC2

Instances

Launch an instance

Key pair name - required

ec2_public

Create new key pair

Network settings

VPC - required

vpc-0c66d8e621ff98a01 (project-vpc)

Subnet

subnet-0ed121c900a1d8efc project-subnet-public2-us-east-1b

Auto-assign public IP

Disable

Firewall (security groups)

Create security group

Select existing security group

my-sql sg-0cac605cbe39c9d91

Compare security group rules

Advanced network configuration

Summary

Number of instances

1

Software image (AMI)

Amazon Linux 2023 AMI 2023.8.2...read more

Virtual server type (instance type)

t3.micro

Firewall (security group)

my-sql

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

Install my-sql client on our public ec2

```
#
##
###
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-172-31-26-67 ~]$ sudo apt update
[sudo] apt: command not found
[ec2-user@ip-172-31-26-67 ~]$ sudo dnf update
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-26-67 ~]$ sudo wget https://dev.mysql.com/get/mysql80-community-release-el9-4.noarch.rpm
--2025-09-22 20:43:18-- https://dev.mysql.com/get/mysql80-community-release-el9-4.noarch.rpm
Resolving dev.mysql.com (dev.mysql.com)... 23.1.40.99, 2000:1400:c400:1201::203, 2000:1400:c400:1801::1203
Connecting to dev.mysql.com (dev.mysql.com)|23.1.40.99|:443... connected.
HTTP request sent, awaiting response... 302 Moved Temporarily
Location: https://repo.mysql.com/mysql80-community-release-el9-4.noarch.rpm [following]
--2025-09-22 20:43:11-- https://repo.mysql.com/mysql80-community-release-el9-4.noarch.rpm
Resolving repo.mysql.com (repo.mysql.com)... 184.26.42.21, 2000:1400:c400:1602::1602, 2000:1400:c400:160d::160d
Connecting to repo.mysql.com (repo.mysql.com)|184.26.42.21|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 13319 (13K) [application/x-rpm]
Saving to: 'mysql80-community-release-el9-4.noarch.rpm'

mysql80-community-release-el9-4.noarch.rpm 100%[=====] 13.01K --.-KB/s in 0s

2025-09-22 20:43:11 (354 MB/s) - 'mysql80-community-release-el9-4.noarch.rpm' saved [13319/13319]

[ec2-user@ip-172-31-26-67 ~]$ sudo dnf install mysql80-community-release-el9-4.noarch.rpm
Last metadata expiration check: 0:00:22 ago on Mon Sep 22 20:42:53 2025.
Dependencies resolved.

Package Architecture Version Repository Size
-----
Installing:
mysql80-community-release noarch el9-5 @commandLine 13 k

Transaction Summary
-----
Install 1 Package

Total size: 13 k
Installed size: 13 k
Is this ok [y/N]: y
Downloading Packages:
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing:
Installing: mysql80-community-release-el9-5.noarch 1/1
Verifying: mysql80-community-release-el9-5.noarch 1/1
Installed:
mysql80-community-release-el9-5.noarch

Complete!
[ec2-user@ip-172-31-26-67 ~]$
```

Creating a private ec2 with the sg my-sql-server

Key pair name - required
ec2-private-a2 [Create new key pair](#)

▼ Network settings [Info](#)

VPC - required [Info](#)
vpc-0c6d8e621f998a01 (project-vpc) [Create new vpc](#)

Subnet [Info](#)
subnet-0138d466187501f01 project-subnet-private2-us-east-1b [Create new subnet](#)
VPC: vpc-0c6d8e621f998a01 Owner: 989567328520
Availability Zone: us-east-1b (use1-aaz2) Zone type: Availability Zone
IP addresses available: 4091 CIDR: 10.0.144.0/20 Subnet: 0138d466187501f01

Auto-assign public IP [Info](#)
Disable

Firewall (security groups) [Info](#)
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
☐ Create security group ☒ Select existing security group

Common security groups [Info](#)
Select security groups
my-sql-server sg-0e41b7998d03e4d4b [Compare security group rules](#)
VPC: vpc-0c6d8e621f998a01

Security groups that you add or remove here will be added to or removed from all your network interfaces.

► Advanced network configuration

▼ Summary
Number of instances [Info](#)
1

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-036dc520857e3138f

Virtual server type (instance type)
t3.micro

Firewall (security group)
my-sql-server

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

SSH into public IP and installing my-sql client in the private

```
Last login: Mon Sep 22 16:10:16 on tty010
(base) lsysar@Lsysar-MacBook-Air:~$ M chmod 400 "ec2_public.pem"
chmod: ec2_public.pem: No such file or directory
(base) lsysar@Lsysar-MacBook-Air:~$ M cd Downloads
(base) lsysar@Lsysar-MacBook-Air:Downloads$ M chmod 400 "ec2_public.pem"
chmod: ec2_public.pem: No such file or directory
(base) lsysar@Lsysar-MacBook-Air:Downloads$ M cd ~
zsh: command not found: ec2-3-94-183-198.compute-1.amazonaws.com
(base) lsysar@Lsysar-MacBook-Air:Downloads$ M ssh -i "ec2_public.pem" ec2-user@ec2-3-94-183-198.compute-1.amazonaws.com
The authenticity of host 'ec2-3-94-183-198.compute-1.amazonaws.com (3.94.183.198)' can't be established.
ED25519 key fingerprint is SHA256:feyjyVHwZ2igdcVU92f0c1hmk10AeadrJNG5MS.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-3-94-183-198.compute-1.amazonaws.com' (ED25519) to the list of known hosts.

#
##
###
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-172-31-26-67 ~]$
```

Installing db on private EC2

```
Running scriptlet: mariadb105-server-utils-3:10.5.29-1.amzn2023.0.1.x86_64
Verifying : mariadb-connector-c-3:3.10-1.amzn2023.0.1.x86_64
Verifying : mariadb-connector-c-config-3:3.10-1.amzn2023.0.1.noarch
Verifying : mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64
Verifying : mariadb105-backup-3:10.5.29-1.amzn2023.0.1.x86_64
Verifying : mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64
Verifying : mariadb105-cracklib-password-check-3:10.5.29-1.amzn2023.0.1.x86_64
Verifying : mariadb105-errmsg-3:10.5.29-1.amzn2023.0.1.x86_64
Verifying : mariadb105-gssapi-server-3:10.5.29-1.amzn2023.0.1.x86_64
Verifying : mariadb105-server-3:10.5.29-1.amzn2023.0.1.x86_64
Verifying : mariadb105-server-utils-3:10.5.29-1.amzn2023.0.1.x86_64
Verifying : mysql-libs-1.0.4-2.amzn2023.0.3.noarch
Verifying : perl-1.88-477.amzn2023.0.7.x86_64
Verifying : perl-DBD-MariaDB-1.22-1.amzn2023.0.4.x86_64
Verifying : perl-DBI-1.643-7.amzn2023.0.3.x86_64
Verifying : perl-Data-Dumper-2.174-468.amzn2023.0.2.x86_64
Verifying : perl-File-Copy-2.34-477.amzn2023.0.7.noarch
Verifying : perl-FileHandle-2.83-477.amzn2023.0.7.noarch
Verifying : perl-File-Bigint-1.11.9998.39-2.amzn2023.0.2.noarch
Verifying : perl-File-Bigint-0.2024-588.amzn2023.0.2.noarch
Verifying : perl-File-Complex-1.69-477.amzn2023.0.7.noarch
Verifying : perl-Sys-Hostname-1.23-477.amzn2023.0.7.x86_64
Verifying : perl-base-2.27-477.amzn2023.0.7.noarch

Installed:
mariadb-connector-c-3:3.10-1.amzn2023.0.1.x86_64
mariadb105-backup-3:10.5.29-1.amzn2023.0.1.x86_64
mariadb105-errmsg-3:10.5.29-1.amzn2023.0.1.x86_64
mariadb105-server-utils-3:10.5.29-1.amzn2023.0.1.x86_64
perl-DBD-MariaDB-1.22-1.amzn2023.0.4.x86_64
perl-File-Copy-2.34-477.amzn2023.0.7.noarch
perl-File-Bigint-0.2024-588.amzn2023.0.2.noarch
perl-base-2.27-477.amzn2023.0.7.noarch

mariadb-connector-c-config-3:3.10-1.amzn2023.0.1.noarch
mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64
mariadb105-gssapi-server-3:10.5.29-1.amzn2023.0.1.x86_64
mysql-libs-1.0.4-2.amzn2023.0.3.noarch
perl-DBI-1.643-7.amzn2023.0.3.x86_64
perl-FileHandle-2.83-477.amzn2023.0.7.noarch
perl-File-Complex-1.69-477.amzn2023.0.7.noarch
perl-Sys-Hostname-1.23-477.amzn2023.0.7.x86_64
perl-base-2.27-477.amzn2023.0.7.noarch

mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64
mariadb105-cracklib-password-check-3:10.5.29-1.amzn2023.0.1.x86_64
mariadb105-server-3:10.5.29-1.amzn2023.0.1.x86_64
perl-1.88-477.amzn2023.0.7.x86_64
perl-Data-Dumper-2.174-468.amzn2023.0.2.x86_64
perl-File-Bigint-1.11.9998.39-2.amzn2023.0.2.noarch
perl-Sys-Hostname-1.23-477.amzn2023.0.7.x86_64

Complete!
[ec2-user@ip-10-0-150-249 ~]$ sudo systemctl start mariadb
[ec2-user@ip-10-0-150-249 ~]$ sudo systemctl enable mariadb
Created symlink /etc/systemd/system/mysql.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/mysql.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/multi-user.target.wants/mariadb.service → /usr/lib/systemd/system/mariadb.service.
[ec2-user@ip-10-0-150-249 ~]$ sudo systemctl status mariadb
● mariadb.service - MariaDB 10.5 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: disabled)
   Active: active (running) since Mon 2025-09-22 21:48:43 UTC; 28s ago
     Docs: man:mariadb(8)
           https://mariadb.com/kb/en/library/systemd/
   Main PID: 16913 (mariadb)
   Status: "Taking your SQL requests now..."
   Tasks: 14 (limit: 1657)
   Memory: 66.4M
   CPU: 58ms
   CGroup: /system.slice/mariadb.service
           └─023 /usr/libexec/mariadbd --basedir=/usr

Sep 22 21:48:43 ip-10-0-150-249.ec2.internal mariadb-prepare-db-dir[16332]: The second is mysql.localhost, it has no password either, but
Sep 22 21:48:43 ip-10-0-150-249.ec2.internal mariadb-prepare-db-dir[16332]: you need to be the system 'mysql' user to connect.
Sep 22 21:48:43 ip-10-0-150-249.ec2.internal mariadb-prepare-db-dir[16332]: After connecting you can set the password, if you would need to be
Sep 22 21:48:43 ip-10-0-150-249.ec2.internal mariadb-prepare-db-dir[16332]: able to connect as any of these users with a password and without sudo
Sep 22 21:48:43 ip-10-0-150-249.ec2.internal mariadb-prepare-db-dir[16332]: See the MariaDB Knowledgebase at https://mariadb.com/kb
Sep 22 21:48:43 ip-10-0-150-249.ec2.internal mariadb-prepare-db-dir[16332]: Please report any problems at https://mariadb.org/jira
Sep 22 21:48:43 ip-10-0-150-249.ec2.internal mariadb-prepare-db-dir[16332]: The latest information about MariaDB is available at https://mariadb.org/.
Sep 22 21:48:43 ip-10-0-150-249.ec2.internal mariadb-prepare-db-dir[16332]: Consider joining MariaDB's strong and vibrant community:
Sep 22 21:48:43 ip-10-0-150-249.ec2.internal mariadb-prepare-db-dir[16332]: https://mariadb.org/get-involved/
Sep 22 21:48:43 ip-10-0-150-249.ec2.internal systemd[1]: Started mariadb.service - MariaDB 10.5 database server.
[ec2-user@ip-10-0-150-249 ~]$
```

Running commands to create the database

```
Connection to 10.0.150.249 closed.
[ec2-user@ip-10-0-27-37 ~]$ ssh -i "key.pem" ec2-user@10.0.150.249
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Mon Sep 22 21:45:11 2025 from 10.0.29.37
[ec2-user@ip-10-0-150-249 ~]$ sudo mysql
Welcome to the MariaDB monitor. Commands end with ; or \g.
Your MariaDB connection id is 3
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> ALTER USER 'root'@'localhost' IDENTIFIED BY 'Jp0krljfgd5e';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.000 sec)

MariaDB [(none)]> CREATE DATABASE testdb;
Query OK, 1 row affected (0.000 sec)

MariaDB [(none)]> CREATE USER 'testuser'@'%' IDENTIFIED BY 'Jp0krljfgd5e';
Query OK, 0 rows affected (0.002 sec)

MariaDB [(none)]> GRANT ALL ON testdb.* TO 'testuser'@'%';
Query OK, 0 rows affected (0.002 sec)

MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.001 sec)
```

Testing the db from public ec2

```
[ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use near 'sudo vi /etc/my.cnf.d/mariadb-server.cnf' at line 1
MariaDB [(none)]> exit
Bye
[ec2-user@ip-10-0-150-249 ~]$ sudo dnf install -y mariadb105
Last metadata expiration check: 8:08:11 ago on Mon Sep 22 21:45:59 2025.
Package mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-10-0-150-249 ~]$ mysql -h 10.0.150.249 -u testuser -p testdb
Enter password:
Welcome to the MariaDB monitor. Commands end with ; or \g.
Your MariaDB connection id is 6
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [testdb]> █
```